

Literature Review & Data Collection

Project: The Revolution and Limitations of Proof-of-Stake: A Study of Blockchain Consensus Mechanisms

Owner: Khải (Literature Review & Data Collection)

Goal: Complete the literature review - establish a solid foundation on PoW/PoS and secure credible academic sources for the three limitation topics the group will analyze in Week 3 (long-range attacks, weak subjectivity, validator centralizationG).

1. Foundational sources

#	Source	Notes on usage
1	Nakamoto, <i>Bitcoin: A Peer-to-Peer Electronic Cash System</i> (2008)	Foundation of PoW - longest-chain rule, proof-of-work puzzle
2	Buterin, <i>Ethereum Whitepaper</i> (2014)	Motivation for a smart-contract platform; precursor to the shift toward PoS
3	Castro & Liskov, <i>Practical Byzantine Fault Tolerance</i> (1999)	Theoretical foundation of BFT - used to compare against PoS in Week 3
4	Pass & Shi, <i>The Sleepy Consensus</i> (2017)	Formal model for consensus with intermittently-online participants - relevant to validator liveness

2. Newly collected sources

2.1 Weak Subjectivity

- Buterin, "**Proof of Stake: How I Learned to Love Weak Subjectivity**" (Ethereum Foundation Blog, 2014) - the original source of the concept. Buterin argues that PoS cannot be fully "objective" the way PoW is (where the correct state is always the chain

with the most proof-of-work); PoS requires a newly joining node to trust a recent checkpoint to determine the valid chain. → **Primary source.**

- **"STAKESURE: Proof of Stake Mechanisms with Strong Cryptoeconomic Safety"** (arXiv:2401.05797, 2024) - a formal analysis of how Ethereum/Tendermint handle weak subjectivity via a weak-subjectivity period; shows that validators who have already exited staking can still attack because there is nothing left to slash. → use for the "why this problem is not fully solved" section.
- **Asgaonkar, "Weak Subjectivity in Ethereum 2.0"** (2020, ethresear.ch) - a technical explanation of how Ethereum computes the weak subjectivity period.

2.2 Long-Range Attack

- **"A Survey on Long-Range Attacks for Proof of Stake Protocols"** (IEEE, 2019) - a systematic literature review covering all variants of long-range attacks (simple, posterior corruption) and mitigation strategies. → **Primary source for the conceptual section.** (A free full-text PDF is also available on ResearchGate - use that version if library access isn't available)
- **Gaži, "Stake-Bleeding Attacks on Proof-of-Stake Blockchains"** (IACR ePrint 2018/248) - introduces a specific attack variant (stake-bleeding) with quantitative analysis using real Bitcoin data. → use as a concrete, data-backed example.
- **Azouvi et al., "Winkle: Foiling Long-Range Attacks in Proof-of-Stake Systems"** (IACR ePrint 2019/1440) - proposes a technical mitigation, useful for the "future directions" section (Objective 5).

2.3 Validator Centralization

- **Buterin, Herman, Reijnders, Leonardos, "Ethereum Proof-of-Stake under the Microscope"** (arXiv:2005.09817, 2020; extended version published as "Ethereum Proof-of-Stake under Scrutiny," ACM SAC 2023) - an empirical analysis of centralization risks in Ethereum's PoS design. → **Primary source, authored by the Ethereum PoS design team, high credibility.**
- **Grandjean, Heimbach, Wattenhofer, "Ethereum Proof-of-Stake Consensus Layer: Participation and Decentralization"** (arXiv:2306.10777, 2023) - post-Merge (2022) empirical data measuring actual validator-set concentration. → more current than the original whitepapers, good for real-world figures.
- **Kiayias, Koutsoupias, Marmolejo-Cossio, Stouka, "Balancing Participation and Decentralization in Proof-of-Stake Cryptocurrencies"** (arXiv:2407.08686, 2024) - a game-theoretic model of the trade-off between incentivizing participation (via stake pools/delegation) and maintaining decentralization. → useful for analyzing root causes rather than just symptoms.
- **Kuśmierz & Overko, "How Centralized is Decentralized? Comparison of Wealth Distribution in Coins and Tokens"** (arXiv:2207.01340, 2022) - Gini-coefficient data comparing wealth concentration across blockchains. → quantitative support for the "wealth concentration" issue mentioned in the Project Description.

2.4 Foundational PoS mechanism sources (for Hiếu & Long — Week 2 "Study Proof of Stake")

These sources explain **how PoS actually works** (staking, validator selection, slashing, protocol design) - separate from the limitation sources above, which explain what can go **wrong** with it.

- **Kiayias, Russell, David, Oliynykov, "Ouroboros: A Provably Secure Proof-of-Stake Blockchain Protocol"** (CRYPTO 2017) - the first PoS protocol with formal, rigorous security proofs (comparable rigor to Bitcoin's security analysis). Explains epochs, slots, random slot-leader election ("Follow the Satoshi"), and a reward mechanism that makes honest behavior a Nash equilibrium. → **Best starting point for "how does PoS actually guarantee security" - foundational, peer-reviewed (Springer).**
- **Buterin & Griffith, "Casper the Friendly Finality Gadget"** (arXiv:1710.09437, 2017) - introduces Casper FFG, the finality mechanism Ethereum uses on top of its chain. Explains checkpoints, justification/finalization, and how validators get penalized (slashed) for violations like double-voting. → **Directly explains Ethereum's actual slashing conditions - core reading for "slashing" and "protocol design."**
- **Ethereum Foundation, "The Beacon Chain" - Ethereum Consensus Specs** (official documentation, ethereum.github.io/consensus-specs) - the official technical spec: how validators register (deposit ETH), get activated, are randomly selected to propose/attest each slot, and get slashed (four specific slashing conditions: double proposal, double vote, surround vote, etc.). → **Official primary source - most concrete and up to date, good for exact mechanics.**
- **"Upgrading Ethereum" (eth2book.info) - One Page Annotated Spec** - a plain-English annotated walkthrough of the Beacon Chain spec above (slots, epochs, committees, validator lifecycle).

3. Quick-reference table

Topic	Key source	Source type	Reading priority
Weak subjectivity	Buterin (2014) original blog post	Primary/blog	High
Weak subjectivity	STAKESURE (arXiv 2024)	Peer-reviewed/preprint	Medium
Long-range attack	IEEE Survey (2019)	Survey	High
Long-range attack	Stake-Bleeding (IACR 2018)	Preprint	Medium

Topic	Key source	Source type	Reading priority
Validator centralization	Buterin et al., "Under the Microscope" (2020/2023)	Peer-reviewed (ACM SAC)	High
Validator centralization	Grandjean et al. (2023)	Preprint, new empirical data	High
Validator centralization	Kiayias et al. (2024)	Game theory, preprint	Low–Medium (advanced)

4. Notes

Some of the sources above are preprints - papers posted online (arXiv, IACR ePrint) that have not yet been through official peer review. They may still be good sources, but they haven't been checked by other experts yet, so they carry less weight than a peer-reviewed paper.

Action item for the reference list (Long - Documentation & Report Formatting): when compiling the final reference list, mark which sources are preprints. This way the whole group knows, and we avoid losing points for citing unverified sources without disclosing it.

Note: "Ethereum Proof-of-Stake under the Microscope" also has a peer-reviewed version, published at ACM SAC 2023 under the title "Ethereum Proof-of-Stake under Scrutiny" - use that version in the final citation instead of the arXiv preprint.